

MOHAMMAD BILAL AZIZ

SOFTWARE DEVELOPER

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EDUCATION

Master's of Science, Artificial Intelligence

Currently Enrolled

- Friedrich-Alexander-Universität Erlangen-Nürnberg

Bachelor's of Science, Computer Science

Aug 2020 - June 2024

- National University of Computer and Emerging Technologies
- CGPA: 3.93/4.0 (Bronze Medal)

TECHNICAL SKILLS

Python, Flask, Fast API

JavaScript, Angular

SQL, Postgres, Redis, Supabase

AWS EC2, AWS Neptune

Vue, React, Next

Claude Code, Cursor, CoPilot

ElasticSearch, EFK

Vuetify, Tailwind, Ng-zorro

Dockers, Kubernetes, GCP

PROFESSIONAL EXPERIENCE

SECURITI.AI : SOFTWARE DEVELOPER

June 2024 - Present

- Made use of cutting-edge **AI agents** to boost the productivity of development by **3x**. Used tools like **Claude code**, **Cursor** and **Github Copilot** to create an extension that resolved over **70+ high-priority accessibility issues** within **8 weeks** by engineering custom accessible wrappers around **Vuetify** components in Vue.js, ensuring full **WCAG 2.2** compliance.
- Led the development and maintenance of **Securiti's Graph Explorer**, a core feature of the platform, empowering users to explore complex data relationships. Utilized advanced **Vue.js** and **Vuetify** patterns alongside the **yFiles (yWorks)** library to implement interactive graph modes and a custom query builder.
- Redesigned graph processing architecture to scale to **10 million** records by integrating **DuckDB** as a persistent cache layer. This solution offloaded heavy graph queries from **AWS Neptune** and **Elasticsearch**, boosting performance and reducing latency for multi-tenant environments. Deployed using **Docker** and managed via **Kubernetes** for seamless scalability and high availability.

SWAAPT.Y: FULL STACK DEVELOPER

Freelance

- Built a full-stack **MVP** for a barter-based eCommerce platform using **Next.js** (App Router), **Supabase** (PostgreSQL, Auth), and **TailwindCSS**, with a fully responsive UI and real-time listings. Designed a scalable product schema and integrated public/private user profiles for dynamic item visibility.
- Implemented **server-side rendering**, dynamic routing, and **RESTful API routes**, optimizing performance and ensuring **SEO** readiness.
- Also set up CI/CD deployment pipeline via **Jenkins**, enabling automated testing and **continuous integration** on staging environments.

FAININ: ANGULAR INTERN

November 2023 - February 2023

- Integrated **OpenAI's API** into the product interface, enabling users to auto-generate high-quality product descriptions and configure workflows with minimal input. This significantly improved user efficiency, reduced manual effort, and introduced AI-driven intelligence to the user journey.
- Implemented the core **inventory management feature** to streamline user workflows and enhance overall usability. Leveraged **Angular's Ng-zorro** library and UX principles to deliver an intuitive, responsive experience that raised the standard for operational efficiency and reduced task completion time.
- Drove **end-to-end project delivery** through structured product management using **Jira**, tracking sprints, issues, and progress across cross-functional teams. Maintained clean, version-controlled codebases using **Bitbucket** and **Git** best practices, fostering collaboration, scalability, and code quality across deployments.

RESEARCH

August 2024

zkVML: Zero Knowledge Verifiable Machine Learning

- Presented the paper at the prestigious **APAN' 58 International Conference** held at Islamabad, Pakistan.
- Designed and proposed novel python framework to create secure, verifiable machine learning models.